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# Accelerating Big Data Analysis through LASSO-Random Forest Algorithm in QSAR Studies

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## Abstract

**Motivation:** The aim of quantitative structure–activity prediction (QSAR) studies is to identify novel drug-like molecules that can be suggested as lead compounds by means of two approaches, which are discussed in this article. First, to identify appropriate molecular descriptors by focusing on one feature-selection algorithms; and second to predict the biological activities of designed compounds. Recent studies have shown increased interest in the prediction of a huge number of molecules, known as Big Data, using deep learning models. However, despite all these efforts to solve critical challenges in QSAR models, such as over-fitting, massive processing procedures, is major shortcomings of deep learning models. Hence, finding the most effective molecular descriptors in the shortest possible time is an ongoing task. One of the successful methods to speed up the extraction of the best features from big datasets is the use of least absolute shrinkage and selection operator (LASSO). This algorithm is a regression model that selects a subset of molecular descriptors with the aim of enhancing prediction accuracy and interpretability because of removing inappropriate and irrelevant features.

**Results:** To implement and test our proposed model, a random forest was built to predict the molecular activities of Kaggle competition compounds. Finally, the prediction results and computation time of the suggested model were compared with the other well-known algorithms, i.e. Boruta-random forest, deep random forest and deep belief network model. The results revealed that improving output correlation through LASSO-random forest leads to appreciably reduced implementation time and model complexity, while maintaining accuracy of the predictions.

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**Supplementary information:** Supplementary data are available at *Bioinformatics* online.

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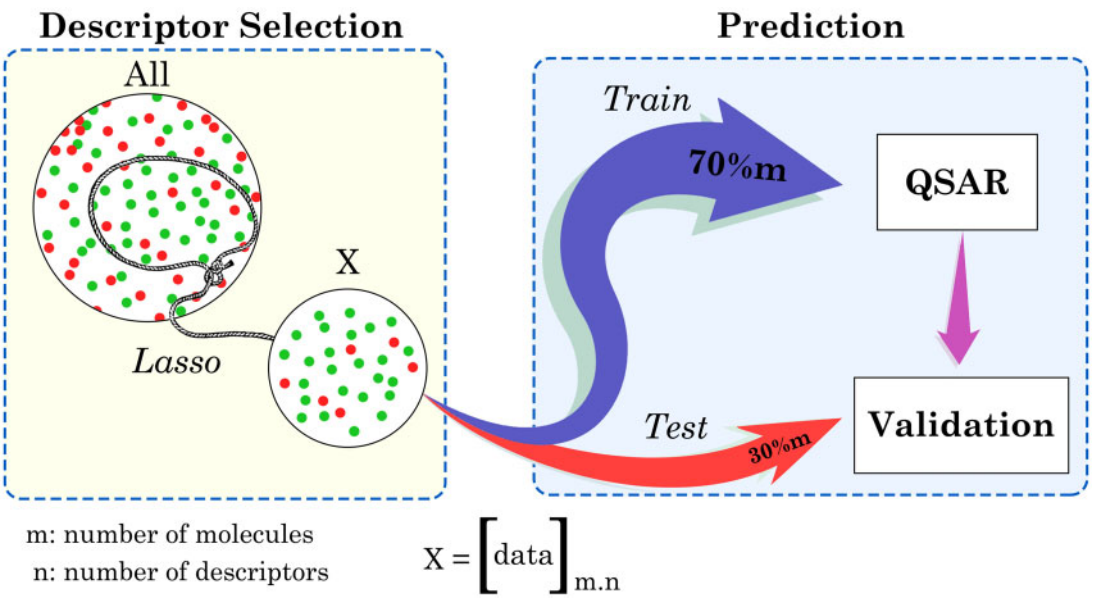


## 1 Introduction

The first quantitative structure–activity relationship (QSAR) equation, which was a milestone in medicinal chemistry, was developed in 1962 by Hansch, one of the pioneers in computer-assisted molecule design (Cherkasov *et al.*, 2014). Mathematical and statistical QSAR models have proved to be two of the most effective computational approaches in *de novo* drug design, since they entail lower expenditure and complexity compared with experimental synthesis. Generally, the main inputs in QSAR models are based on the numerical information provided by what are known as molecular descriptors, which play a fundamental role in pharmaceutical science and describe the physicochemical features and 0D–4D fingerprints of drug-like compounds (Todeschini and Consonni, 2008). Recently, new software and tools have enabled scientists to compute many molecular descriptors at relatively little cost.

Big Data in drug discovery refers to processing huge number of potential drug-like compounds, both in academia and the pharmaceutical industry, as well as in numerous molecular descriptors. Over the last decade, Big Data has been increasingly gaining the attention of researchers as they attempt to solve critical challenges in QSAR models, including redundancy, getting stuck in local minima and over-fitting (AlNuaimi *et al.*, 2019). As a new research paradigm, several deep learning architectures have been developed as an alternative to shallow learning models and have become popular tools in computational drug design as a possible solution for capturing complex statistical patterns among thousands of descriptors extracted from numerous compounds since 2015. Deep learning architectures contain multiple levels of linear and non-linear operations with a hierarchical construction in which higher-level features are calculated from lower-level ones. The depth of deep learning models refers to the longest path between an input and output node (Ghasemi *et al.*, 2018; Sagiroglu, 2013).

Despite improved prediction accuracy and solving over-fitting problems using deep learning algorithms, their major shortcomings are that they are time consuming and the required massive processing procedure (Daniel, 2019; Eklund *et al.*, 2014). In addition, there is a tradeoff between finding non-linear patterns of features and model complexity in Big Data processing, e.g. computational drug design, digital signal processing and image processing. Data mining techniques provide solutions with regard to gleaning the most essential information from input data through data mapping in another space or exploring the most effective subset of features. For this reason, selecting an appropriate data mining approach is vital when the model input contains redundant and irrelevant information; in this way, the risk of the above-mentioned challenges should be reduced (Arian *et al.*, 2020).

Different feature selection methods, including input models and optimized prediction algorithms, have been proposed for use in massive datasets to find the most informative features. Three different data mining categories have been proposed—filter, wrapper and embedded. Filter methods are used to reduce the dimensions of input data, regardless of experimental values, and are only based on the intrinsic properties of the data. Simplicity, increased speed and a reduction in computational time count are among the advantages of the filter method, while its lack of interaction with classification/regression is its main disadvantage. Wrapper techniques are used to select the optimum subset of features based on the error reduction of classifier/prediction algorithms. These methods perform better than filter methods even though they are more expensive and time consuming (Jurtz *et al.*, 2017; Saeys and Inza, 2007).

However, despite the advantages of both approaches, each of them has drawbacks when dealing with complex data, and for this reason, it is quite common to merge the filter and wrapper approaches, creating a hybrid/embedded approach, which retains the advantages of both approaches, while reducing their drawbacks (AlNuaimi *et al.*, 2019; Xiao *et al.*, 2018).

In this respect, an attempt was made to improve biological activity prediction of drug-like compounds in the QSAR model by using Kaggle competition datasets (sponsored by Merck in 2012), which involved implementing deep learning algorithms for the very first time in drugs (Ghasemi *et al.*, 2018). In the largest Kaggle datasets, there are more than 50 000 molecules and 8700 descriptors for each compound. One of the main features of the data is the existence of a huge number of compounds and various redundant and irrelevant descriptors, along with a non-linear relationship between them that should be discarded. A wealth of valuable research has been published in leading journals since 2015 and have focused on deep learning techniques (Jing *et al.*, 2018; Kursa *et al.*, 2010; Ludwig *et al.*, 2015; Su *et al.*, 2019; Xu *et al.*, 2019).

As mentioned above, the running time and massive processing required are the main drawbacks of deep learning models, and, as consequence, several algorithms have been developed to reduce both. Among them, Boruta, multivariate adaptive regression splines (MARS) and Relief are some of the most widely used feature-selection algorithm in computational drug design as it provides criteria for extracting the most important descriptors (Algamal *et al.*, 2015; Jurtz *et al.*, 2017). In 2014, MARS, Relief and elastic nets,

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together with eight learners were considered on five real datasets. In conclusion, it was shown that choosing well-suited feature selection methods in combination random forest could improve model transparence. Besides, at the end of conclusion, it was emphasized that learner, e.g. RF, without feature selection, did not handle many features, compared to the number of observations (Saeys and Inza, 2007). As part of Boruta research, two datasets, GDSC and CCLE, were used to predict molecular IC50 values. However, it should be noted that in QSAR studies, the time required to complete a Boruta run on a personal computer for Big Data is quite high (Xia *et al.*, 2018). Therefore, apart from selecting a subset of relevant descriptors, it seems that a strong high-speed technique is needed for it to be effective for the outputs of the QSAR model applied to Kaggle datasets.

To solve this problem, an embedded model, the Least Absolute Shrinkage and Selection Operator (LASSO), is proposed as an algorithm for descriptor selection. In the LASSO algorithm, a regularization term is added to the estimator to keep the coeffcients of the regression small. Therefore, LASSO implicitly conducts a feature selection, since variables are inherently driven to zero (Nguyen and Huang, 2015; Yamada *et al.*, 2018). LASSO structure computation is concerned with extracting a subset of features to reach a parsimonious description of the system, which may afford greater insight into its functionality (Muthukrishnan, 2016). Finally, random forest algorithm has been used to predict molecular biological activity. Random forest (RF), an ensemble model of decision trees from random subsets of features, has generally been used as a powerful feature selection, classification and prediction method. However, when working with high-dimensional data, RF has poor accuracy (Datta and Dev, 2019).

To achieve the best evaluation, all the compounds of each dataset were clustered into four different categories, according to their biological activity values. After that, 70% of each group was extracted randomly as a training set, and the remaining 30% of molecules were retained for test sets (Fig. 1). These procedures were repeated twenty times, and the results were averaged. It was expected that more accurate estimations and fewer errors would be found as a result of better correlations and lower root mean square error, respectively, with shorter computation times.

## 2 Materials and methods

### 2.1 Feature selection methods

As mentioned, feature selection methods have been divided into three different groups, filter, wrapper and embedded techniques. According to previous publications in computational drug design, LASSO, Boruta, Relief and MARS had the best performances among other approaches in small datasets (Algamal *et al.*, 2015; Jurtz *et al.*, 2017). Therefore, we persuaded to examine types of feature selection in Big data to reduce computational cost and time.

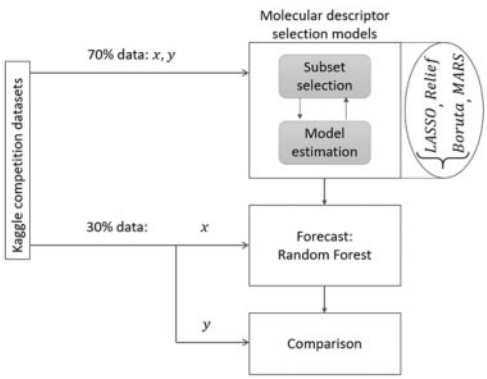


**Fig. 1.** Schematic of the proposed model

#### 2.1.1 Lasso

The huge number of observations and variables persuaded Robert Tibshirani to introduce a new powerful embedded feature selection method in 1996, called least absolute shrinkage and selection operator (LASSO), with the aim of increasing the accuracy of prediction and removing inappropriate and irrelevant features. In the LASSO algorithm, optimal variables are extracted from the datasets by a simple method, i.e. least squares regression with a special penalty term. The following formula describes the LASSO estimation:

$$\hat{\beta} = argmin\left[\frac{1}{2}\sum_{i=1}^{N}\left(y_i - \beta_0 - \sum_{j=1}^{p} x_{ij}\beta_j\right)^2 + \lambda\sum_{j=1}^{p}|\beta_j|\right], \quad (1)$$

where $N$ and $\beta$ are the number of observations in each run and regression coefficient, respectively. Furthermore, to reduce the regression coefficient of the variables and control the rate of shrinkage, $\lambda$ is applied as a penalty in the so-called called regularization process. The main merit of the LASSO algorithm is increasing the prediction accuracy of the model and reducing the variance by applying shrinkage and removal coefficients (Datta and Dev, 2019; Fonti and Belitser, 2017; Kursa *et al.*, 2020; Liu *et al.*, 2015; Zheng and Liu, 2011).

#### 2.1.2 Borura

The Boruta algorithm is one of the most popular wrapper feature selection methods and was introduced by Kursa *et al.* (2010). While other subset selection algorithms attempt to extract the best sub-sets of the feature, based on the decision tree algorithm, Boruta attempts to find the most important variables independently using a binomial distribution. The computed time for running the Boruta algorithm depends on the number of features and observations, meaning that the run-time increases as the number of molecules and molecular descriptors increases (Kumar and Singh, 2021; Xia *et al.*, 2018).

#### 2.1.3 Relief

Relief is one the well-known filter method to extract features from datasets randomly and weighting them based on the k-nearest neighbor algorithm. Two main advantages of Relief are reliability in noisy data and robust to the incomplete or missing data; however, it is slow for large number of observation (Shin *et al.*, 2019).

#### 2.1.4 MARS

MARS is another feature selection method for complex non-linear regression problems. In this method, non-linear relationship between variables is automatically model via extension linear models. Costly process and poorly performance are the main drawbacks of MARS algorithm (Saeys and Inza, 2007).

### 2.2 Biological activity prediction

Random forest, an algorithm based on the decision tree method, includes three sampling techniques: (i) replacement method of the main dataset; (ii) random selection of subsets among the features; (iii) combining the results of all trees. To extract the desired batch of data, the Bootstrap method, which combines the results of the trees using the bag method, is applied to the input data. The parallel structure, one of the most obvious features of a random forest, consists of several, independent decision trees. In this parallel structure, subsets of the samples are generated by the bootstrap technique, and then the decision tree is applied to all these subsets (Fonti and Belitser, 2017; Kensert *et al.*, 2018; Zhou and Feng, 2019).

Deep random forest is inspired from the success of deep convolutional neural networks, which made many scientists wonder whether to use the benefits of deep layered architectures. Indeed, random forest can adaptively deepen the network by increasing the number of layers and trees (Amiri *et al.*, 2016; Zhou and Saghapour, 2021).

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## 3 Results and discussion

### 3.1 Datasets

The proposed model was implemented in the R packages. The database used in this article was downloaded from a Kaggle competition database (www.kaggle.com). Descriptions of each target can be found in Ghasemi *et al.* (2018).

### 3.2 Metrics

To assess the model performance, three different standard metrics were used: squared Pearson correlation coefficient ($R^2$), root mean square error (RMSE) and absolute error (AE).

$$R^2_{y,\hat{y}} = \frac{\left(\sum\left(Y_i - \overline{Y}\right)\left(\hat{Y}_i - \overline{\hat{Y}}\right)\right)^2}{\sum\left(Y_i - \overline{Y}\right)^2 \sum\left(\hat{Y}_i - \overline{\hat{Y}}\right)^2} \quad (2)$$

$$RMSE = \sqrt{\frac{\sum\left(Y_i - \hat{Y}_i\right)^2}{n}} \quad (3)$$

$$AE = \frac{\sum_{i=1}^{n}\left|Y_i - \hat{Y}_i\right|}{n} \quad (4)$$

### 3.3 Parameter selection for training step in proposed models

One of the main problems found in this study was the proposed model's robustness with respect to the different feature selection parameters and how the number of trees affect the algorithm's output. To avoid the explanation of finding hyperparameters of the models in details, we preferred to summarize them in Table 1, and every individual aspect of the model set was discussed in Supporting Information (Supplementary Tables S1–S3).

### 3.4 Investigating the effect of the LASSO algorithm as a feature selection method on the RF results

Since the LASSO algorithm has produced excellent results in other research, we decided to use it as feature selection before applying the random forest model. The selected parameters for LASSO algorithm and random forest were 0.9 penalty and 300 trees, respectively. Then, the root mean squared errors were calculated for all the test sets. Ten different runs were executed with different input data for each Kaggel dataset (see boxplot in Fig. 2). All targets were shown based on the index of the target. As demonstrated with arbitrarily selected parameter settings, the range of RMSE variations was about 0.01 for all the datasets. Moreover, in seven datasets, they were less than 0.001, particularly for the biggest ones (ACT1 and ACT6).

In addition, to evaluate the effect of the LASSO algorithm on the output of the random forest model, the squared Pearson correlation coefficient (R2) was calculated with its standard deviations and *P*-value for all the test sets. As in Figure, 10 different runs were executed with different input data each Kaggel dataset. The results are shown in Table 2 where the mean squared correlations among all datasets are presented at the end of the table.

All targets were displayed based on the index of the target. As can be seen, in most datasets, the squared correlations of LASSO-RF and their confidence intervals were appreciably higher than those obtained by RF without the feature selection technique. In addition, their variances were generally one-tenth of those obtained by RF. Consequently, the results confirm the theory that the correct extraction of descriptors could noticeably improve the various outcomes in computational drug design.

Apart from R2 and RMSE other metrics are useful for evaluating model performance, but as the only criteria, it can be misleading. Absolute error (AE) is other criteria recommended for model validation. As mentioned in the previous section, we would like to compare LASSO performance with the other confirmed feature selection methods in small datasets in QSAR studies. Figure 3 compares the AE and R2, averaged over all datasets ten times (instead of two large datasets, ACT1 and ACT6), and between the five different proposed methods, RF—without a feature selection model—deep RF, Boruta-RF, MARS-RF, Relief-RF and LASSO-RF. The clearest observation from Figure 3 is that in the RF-without feature selection methods-comparison, random forest model as a feature selection algorithm has slightly worse performance. Therefore, it is confirmed that RF is not suitable for feature selection (Saeys and Inza, 2007). Besides, choosing an appropriate feature selection technique is quite challenging and it is related to the features and observations. In contrast, selection of features from a large pool of descriptors via Boruta, MARS, Relief and LASSO led to improve results.

**Table 1.** Overview of the hyperparameter selection to control proposed models

| Model | Parameter | Selected value |
|---|---|---|
| RF | Number of trees | 300 |
| Boruta algorithm | Number of trees | 200 |
| LASSO algorithm | Penalty | 0.9 |

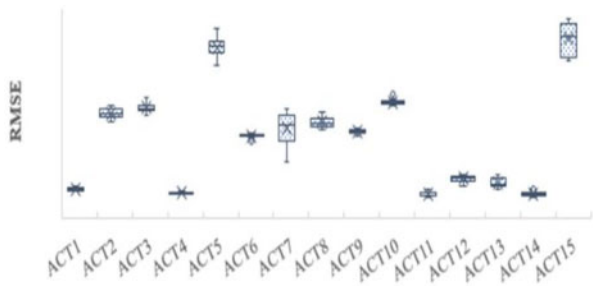


**Fig. 2.** Calculated RMSE for all Kaggle datasets. As shown, the RMSE ranges of majority of them were less than 0.0

**Table 2.** Squared correlation of all targets with their standard deviations for Lasso-RF and RF without feature selection model

| Method | LASSO-RF with recommended parameters | | RF with adjusted parameters | |
|---|---|---|---|---|
| | $R^2 \pm$ SD | *P*-value | $R^2 \pm$ SD | *P*-value |
| ACT1 | 0.678 ± 3e−06 | 0.002 | 0.365 ± 4e−4 | 0.002 |
| ACT2 | 0.672 ± 6 e−05 | 0.01 | 0.654 ± 2e−4 | 0.01 |
| ACT3 | 0.683 ± 6e−05 | 0.012 | 0.665 ± 3e−4 | 0.012 |
| ACT4 | 0.638 ± 3e−4 | 0.001 | 0.414 ± 3e−3 | 0.001 |
| ACT5 | 0.744 ± 7e−05 | 0.015 | 0.704 ± 3e−4 | 0.015 |
| ACT6 | 0.757 ± 2e−06 | 0.002 | 0.50 ± 2e−3 | 0.002 |
| ACT7 | 0.667 ± 3e−4 | 0.03 | 0.40 ± 1e−2 | 0.035 |
| ACT8 | 0.710 ± 2e−05 | 0.012 | 0.68 ± 4e−5 | 0.012 |
| ACT9 | 0.766 ± 1e−05 | 0.001 | 0.70 ± 5e−4 | 0.001 |
| ACT10 | 0.760 ± 1e−05 | 0.005 | 0.706 ± 1e−4 | 0.005 |
| ACT11 | 0.636 ± 1e−4 | 0.002 | 0.508 ± 3e−4 | 0.002 |
| ACT12 | 0.683 ± 4e−05 | 0.002 | 0.496 ± 4e−4 | 0.002 |
| ACT13 | 0.532 ± 2e−4 | 0.002 | 0.431 ± 1e−3 | 0.05 |
| ACT14 | 0.511 ± 1e−4 | 0.001 | 0.385 ± 3e−3 | 0.01 |
| ACT15 | 0.804 ± 1e−05 | 0.019 | 0.754 ± 1e−4 | 0.022 |
| Mean | 0.683 ± 1e−4 | 0.0078 | 0.557 ± 2e−3 | 0.0121 |

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### 3.5 Comparing the proposed model with DBN-DNN, Boruta-RF, MARS-RF and Relief-RF

As mentioned, the careful selection of the best feature selection algorithm could help attain the best prediction. Previously, the results of the five different approaches, ANN, MLR, RF, DNN and DBN-DNN, were published by our research team in 2018, where it was concluded that DBN-DNN had the best performance (Sagiroglu, 2013).

Thus, we decided to compare the proposed model with DBN-DNN and three other well-known methods, Boruta-RF, MARS-RF and Relief-RF (Table 3). In summary, the conclusion of this section is:

1. Choosing an appropriate feature selection method before prediction plays a fundamental role in pre-processing Big-data analysis and have a positive effect on the results.
2. Relief was not be able to perform on two large datasets, ACT1 and ACT6. Thus, it was confirmed that Relief is not appropriate for high-dimensional datasets (Arian *et al.*, 2020).

In addition, regression plot to show the degree of agreement between experimental and predicted data. It should be mentioned that our adjustable parameters were chosen from the optimized groups of results proposed in previous sections. To illustrate the degree of agreement among experimental and predicted data on one of the largest datasets (ACT1), scatter plots of three different models, i.e. MARS-RF, Boruta-RF and LASSO-RF, are shown in Figure 4. As can be seen, the LASSO-RF and Boruta-RF methods agreed with the experimental values better than MARSE-RF.

### 3.6 Comparing the proposed model with DBN-DNN, Boruta-RF, MARS-RF and Relief-RF

With regard to the Big Data challenges mentioned in Section 1, we should mention that the run-time of ACT1 in the comparison with

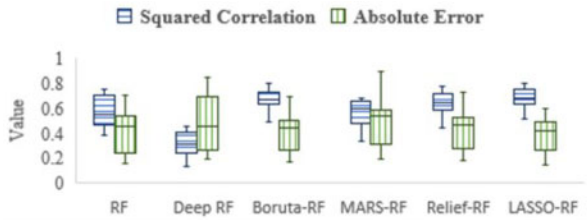


**Fig. 3.** The comparison of AE and squared correlation, averaged on all targets, for the proposed QSAR model with RF-without feature selection, Boruta-RF, deep RF, MARS-RF, LASSO-RF and Relief-RF

**Table 3.** Comparison of the proposed model with some of the well-known feature selection methods

| Dataset index | DBN-DNN | Boruta-RF | MARS-RF | Relief-RF | LASOO-RF |
|---|---|---|---|---|---|
| ACT1 | 0.625 | 0.794 | 0.712 | – | 0.823 |
| ACT2 | 0.857 | 0.819 | 0.775 | 0.803 | 0.820 |
| ACT3 | 0.901 | 0.819 | 0.772 | 0.814 | 0.827 |
| ACT4 | 0.663 | 0.790 | 0.743 | 0.765 | 0.7985 |
| ACT5 | 0.864 | 0.855 | 0.816 | 0.84 | 0.863 |
| ACT6 | 0.774 | 0.824 | 0.784 | – | 0.870 |
| ACT7 | 0.738 | 0.814 | 0.741 | 0.768 | 0.817 |
| ACT8 | 0.860 | 0.841 | 0.808 | 0.803 | 0.843 |
| ACT9 | 0.850 | 0.873 | 0.812 | 0.856 | 0.875 |
| ACT10 | 0.868 | 0.875 | 0.808 | 0.855 | 0.872 |
| ACT11 | 0.712 | 0.802 | 0.685 | 0.764 | 0.798 |
| ACT12 | 0.716 | 0.820 | 0.711 | 0.789 | 0.827 |
| ACT13 | 0.797 | 0.728 | 0.619 | 0.716 | 0.729 |
| ACT14 | 0.654 | 0.702 | 0.585 | 0.662 | 0.715 |
| ACT15 | 0.871 | 0.896 | 0.825 | 0.879 | 0.897 |
| Mean | 0.783 | 0.817 | 0.774 | 0.793 | 0.825 |

Time complexity calculation steps

```
At the top of the algorithm:
T1 <- sys.time() # get start time
      <main codes>
At the end of algorithm:
T2 <- sys.time()-old # calculate difference
Print(T2)
```

Boruta-RF and DBN-DNN, which was previously published by our research team, was a concern at the time. Accordingly, to estimate the run-times, the following codes were written for application at the beginning and end of the algorithms to define time complexity:A surprising observation from Figure 5 was that the running time of the LASSO algorithm was lower than that of other methods. By contrast, when the feature selection algorithm was used for the high-dimensional datasets, the model was slightly efficient than the deep learning algorithm. In addition, compared with the Boruta procedure, the LASSO algorithm was very fast at selecting the most important features, although its average predictive capacity was seen to be approximately the same as the capacity of Boruta-RF.

## 4 Conclusion

One of the main challenges in computational drug design is the existence of a huge number of drug-like compounds, known as Big Data. One of the best-known big datasets is related to the Kaggle competition held in 2012 and sponsored by Merck. According to previously published results, the implementation of a new deep learning algorithm, deep random forest (DRF), was intended to improve the outputs. Despite the potential of RF to predict the biological activities (with a correlation of 0.7833), it is not appropriate for descriptor selection (0.5720 correlation) (Ghasemi *et al.*, 2018).

Since model Run-Time proved to be the main problem in deep learning models, feature selection methods were seen to be by far the best techniques for extracting the relevant subsets of variables. Among the outcomes of various research attempts to find an adequate solution for the mentioned problems, the Boruta, MARS and Relief algorithms were found to be successful in small datasets. Thus, we persuaded to investigate their efficiency in high-dimensional datasets. According to the results, the prediction results of Boruta, MARS, Relief and LASSO had a significantly higher performance as the squared correlation increased and the absolute error reduced (Fig. 3). However, Relief was not executed on two large datasets, ACT1 and ACT6, successfully. Thus, it was confirmed that Relief is not appropriate for high-dimensional datasets. In addition, predicted biological activities with MARS-RF did not agree with the experimental values in Big datasets (Fig. 4); hence the same as Relief, it was concluded that MARS algorithm as feature selection was not suitable for large number of descriptors as well as huge number of molecules.

The execution time of the Boruta algorithm depends closely on the growing number of compounds and molecular descriptors. Precisely for that reason, the LASSO algorithm was proposed. To the best of our knowledge, there is no published research applying the LASSO method in QSAR studies based on the Kaggle datasets as Big data. The Penalty parameter in the LASSO algorithm clearly reduces the model's output at various levels, with no substantial increase in bias. In addition, one important feature of the LASSO algorithm is its ability to select a group of features, whereas the Boruta algorithm attempts to select the features individually. Experimental results showed that, when maintaining the correlation, LASSO-RF could reduce the implementation time of Boruta-RF by up to eight times. It is concluded that a combination of the LASSO algorithm, as a feature selection method, and random forest for prediction, is a novel and efficient solution to the problems mentioned above, reducing both time and complexity.

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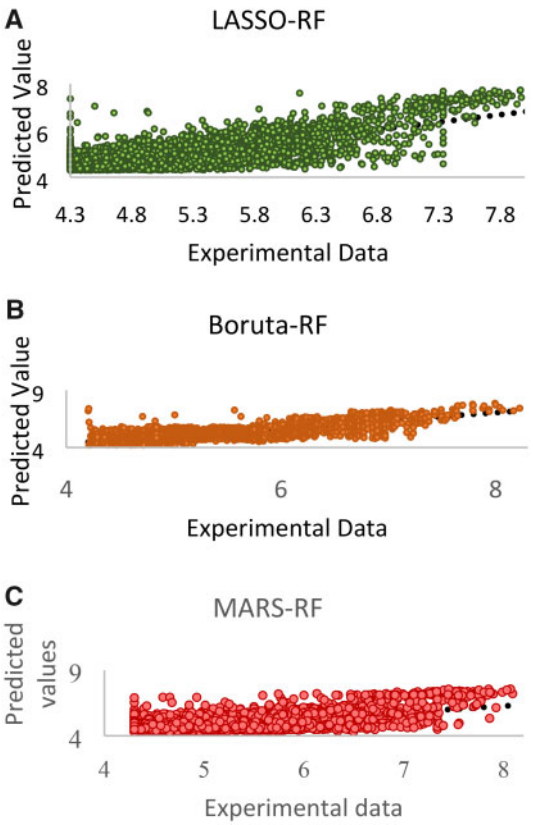


**Fig. 4.** Scatter plot of (**a**) LASSO-RF, (**b**) Boruta-RF and (**c**) MARS-RF. As shown, instead of MARS-RF, two other models have the acceptable correlation between experimental and predicted values

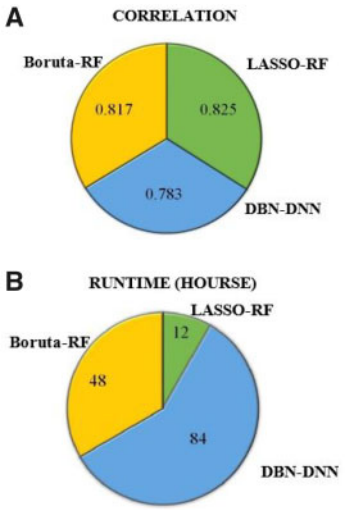


**Fig. 5.** Comparisons of the (**a**) correlation and (**b**) Runtime model, which calculated for ACT1 as a Big Data. It shows that the LASSO algorithm effects on the reduction running time abruptly and increasing prediction power of algorithm

In summary, the conclusions of this study are:

1. Improvement significantly in prediction performance was not observed in using random forest as the feature selection.
2. Improvement in prediction performance with appropriate feature selection could be achieved compared to when no feature selection was used.
3. MARS and Relief performed poorly in high-dimensional datasets.
4. Comparison run time of LASSO-RF with Boruta-RF demonstrates that time saving was significantly occurred by LASSO-RF; furthermore, a high-performance computer could be replaced by personal computer.

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The Kaggle datasets were utilized. This dataset is available on the Internet at: www.kaggle.com.

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*Conflict of Interest*: none declared.

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